

Introduction and housekeeping

Using Networks to Study Microbes

Alberto Santos & Yesid Cuesta-Astroz



Outline

- Who are you?
- Introduction to the course
- Course schedule
- Tools
- Contact

Who are you?

Alberto Santos - Associate Professor and Head of the Informatics Platform at the Danish Technical University (DTU) - Denmark

- Research Group Lead **Multi-omics Network Analytics** (MoNA )

- **Multi-omics**

Using multimodal data to have a comprehensive view on (micro) biology problems

- **Network**

Exploiting graphs to structure, represent, integrate and analyse data

- **Analytics**

Applying Data Science and Machine Learning to answer complex biological questions

Who are you?

Yesid Cuesta-Astroz - Associate professor



- Associate Professor / Universidad de Antioquia / Escuela de Microbiología
 - Pathogen genomics and bioinformatics
 - Working on the study of the biological diversity of microorganisms from omics approaches.
 - Modelling the host-pathogen interactions using protein-protein interaction networks.

Who are you?

Introduction

Introduction to the Course

Networks to Study Microbes

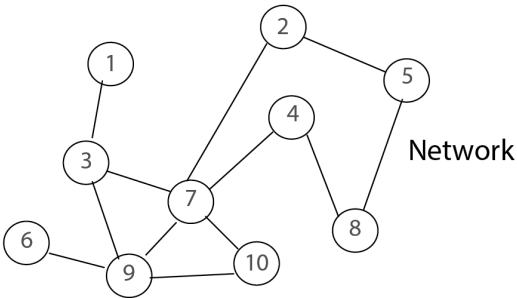
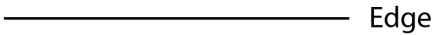
- Technological advances are making **large-scale omics datasets** available to **study microbes and microbial communities**
- We need **computational methods** to **model and understand** these complex data
- This course will provide an overview of **omics data types** and **data resources** available and how to **integrate, analyse and interpret** these data with **networks**

Introduction to the Course

What are networks?

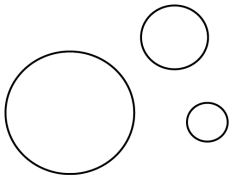
- **Data structures** of components (**nodes**) connected by relationships (**edges**)
- These structures allow:
 - Quick **integration** of **heterogeneous data** based on relationships
 - **Graph theory** methods can be used to **analyse** and **interpret** data, e.g., topological properties can be used to explain:
 - The possible **role** of specific components
 - The **flow** of information
 - The **robustness** of the system
- **Visualize** data

Networks

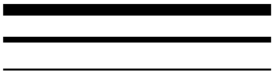


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0	1	1	1	0	0	0	0	1	1
0	0	0	1	1	0	0	0	0	0
0	0	1	0	0	1	1	0	0	1
0	0	0	0	0	0	1	0	1	0

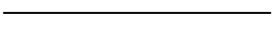
Adjacency matrix



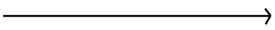
weighted nodes (size)



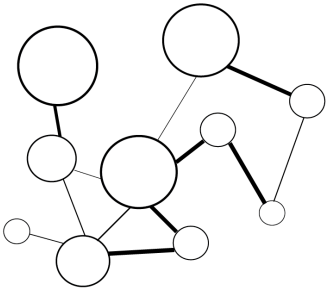
weighted edges (thickness)



undirected edge



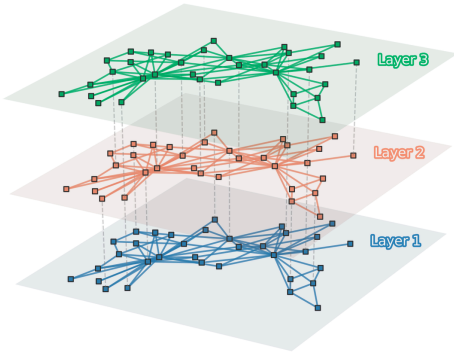
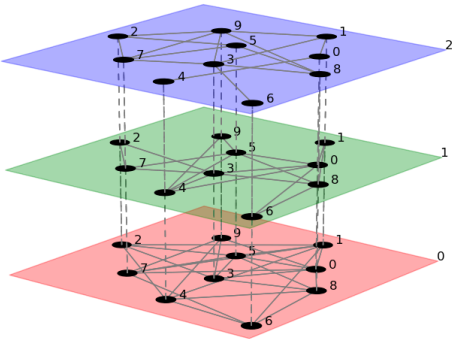
directed edge



weighted undirected network (thickness)

0	0	%	0	0	0	0	0	0	0
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%	0	0	0	0	0	%	0	%	0
0	0	0	0	0	0	%	%	0	0
0	%	0	0	0	0	0	0	%	0
0	0	0	0	0	0	0	0	0	%
0	%	%	%	0	0	0	0	%	%
0	0	%	%	%	0	0	0	0	0
0	0	0	0	0	0	%	0	0	%
0	0	0	0	0	0	0	0	%	0

Weighted adjacency matrix



Schedule

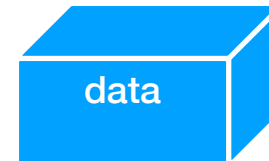
Time	DAY 1	DAY 2	DAY 3	DAY 4
9:00-9:45	Introduction and Housekeeping	Introduction to Python I	Analysing Networks I	Graphical User Interface
9:45-10:30	An Omics View on Microbes I	Introduction to Python II	Analysing Networks II	Graphical User Interface
10:30-11:00	Coffee break	Coffee break	Coffee break	Coffee break
11:00-11:45	An Omics View on Microbes II	Working with Data in Python I	Visualising Networks I	Invited Speaker
11:45-12:30	Open Science	Working with Data in Python II	Network Exercises	Introduction to Team Projects
12:30-14:00	Lunch	Lunch	Lunch	Lunch
14:00-14:45	Network Biology	Databases & Data Annotation	Visualising Networks II - Gephi	Team Project
14:45-15:30	Network Biology journal club	Visualizing Data in Python	Visualising Networks II - Cytoscape	Team Project
15:30-16:00	Coffee break	Coffee break	Coffee break	Coffee break
16:00-16:30	Network Biology journal club	Visualizing Data in Python	Cytoscape/Gephi exercise	Team Project
16:30-17:00	Recap and Q & A	Recap and Q & A	Recap and Q & A	Team Project Presentations and Q&A

Tools

Course Website

<https://github.com/Multiomics-Analytics-Group/networks> to study microbes

- We use GitHub to maintain the contents of the course
- **GitHub** is a framework generally used to maintain and version control software development projects
- **Version control** is a system that records changes to a file or a set of files over time and who made them so that you can recall specific versions later

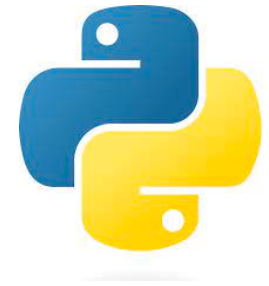


<https://en.wikipedia.org/wiki/GitHub>

<https://git-scm.com/book/en/v2/Getting-Started-About-Version-Control>

Basic Programming

Python



- Python is a **high-level, general-purpose programming language**
- It is a good language for both **beginners** and **advanced programmers**:
 - Its design philosophy **emphasises code readability**
 - Many **learning resources** available
 - **Large community** behind (i.e. stackoverflow for questions)
 - **Excellent Libraries** available, especially for data science

Jupyter Notebooks

<https://jupyter.org/>



- **Web-based development environment** for **creating, running** and **sharing Python** (and other languages) code
- A **notebook** is an interactive document that combines **live code, equations, text or markdown**, and **visualisations** (output of your code)
- Notebooks are divided into **cells** that **run sequentially!** (Need to pay attention)
- It **requires** having **Python installed** on your local machine



Colab Notebooks

<https://research.google.com/colaboratory/faq.html>

- Google Colab is based on **Jupyter Notebook** open source project **hosted on Google's servers**
- Advantages:
 - Requires **no setup** to use (no python installation)
 - Provides **free** access to **computing resources** on Google's servers including GPUs
 - **Notebooks** can be **shared** just as you would with Google Docs or Sheets.
 - You can **import** existing **Jupyter notebooks**
- **Own data and notebooks** need to be accessed through **Google Drive** — Need Google account

Gephi

<https://gephi.org/>



- **Open-source software** for **visualising** and **analysing** large networks graphs
- Used for any kind of networks, especially large networks — good performance
- **Extendable** using Plugins — <https://gephi.org/plugins>

Cytoscape

<https://cytoscape.org/>



- An open source **software platform** for **visualising** and **analysing** complex **networks**
- Used for any kind of networks but **specialised** on **biological domains**:
e.g, Molecular interaction networks and biological pathways and integrating these networks with annotations, gene expression profiles and other state data.
- **Additional features** are available as freely available **Apps** (<https://apps.cytoscape.org/>)

Data Literacy

Data is like garbage. You'd better know what you are going to do with it before you collect it. Mark Twain

What is it?

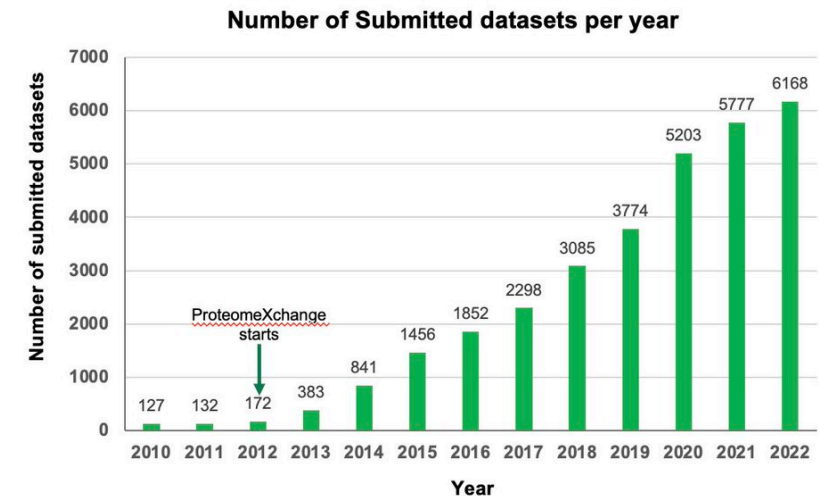
Data as a universal language

- The ability to **read, analyse** and **communicate** data effectively
- Includes also understanding **data sources** and **formats**, and the **analytical methods** to transform them into results
- Requires skills and knowledge in **data analysis, statistics, data visualisation**, and **data management**.
- It does not mean that we all need to be **Data Scientist ...**
- **but** we all need certain skills to extract value from data:
 - **Non-technical**: interpretation, critical thinking, communication, evaluation
 - **Technical**: analysis, visualisation, programming, management

Why?

Data! Data! Data! Evidence to support our research

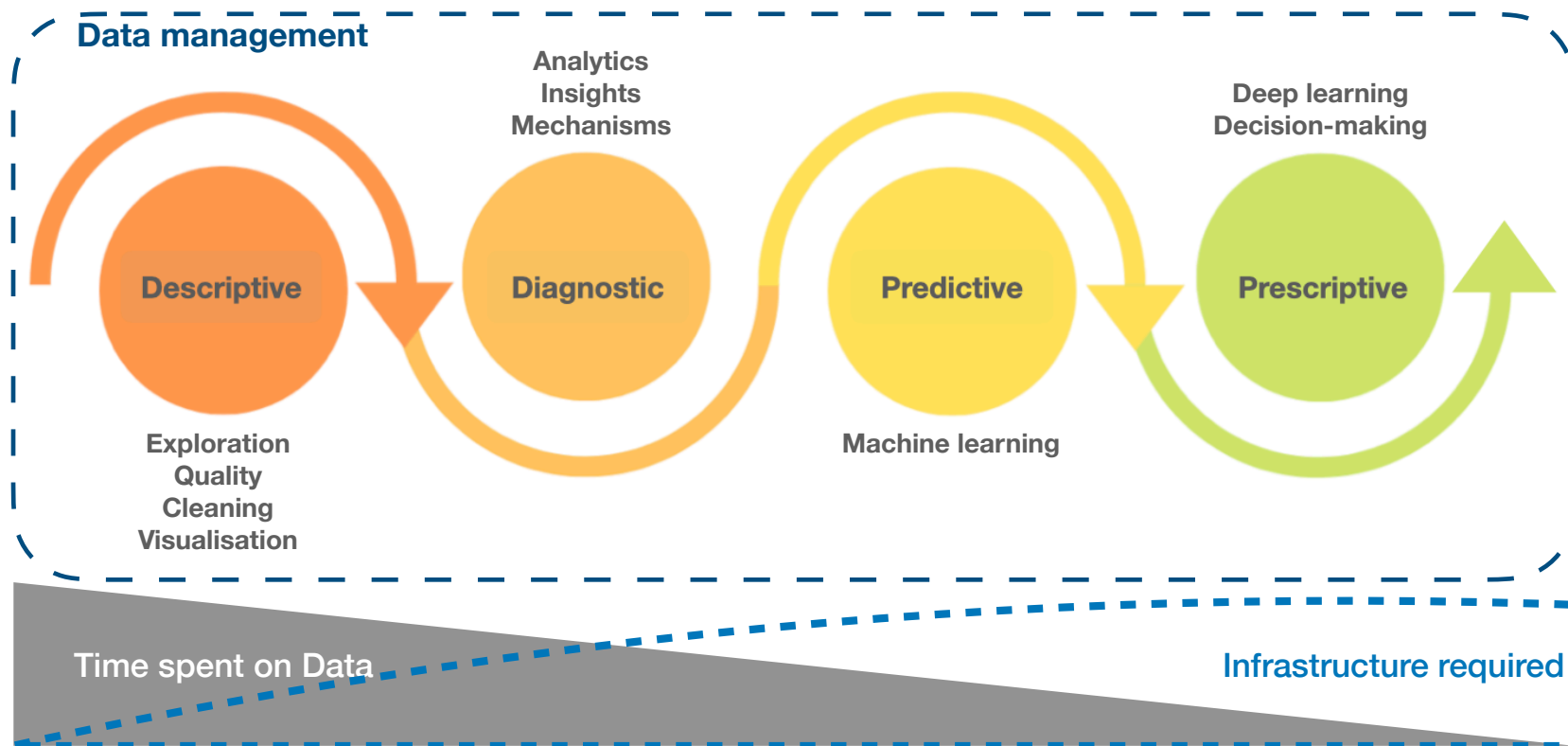
- We **generate data** everyday: metadata, raw, processed, clean, derived results
- **Growing number** of **resources** and **datasets** available
- The data can have **multiple purposes** — reusability (FAIR)
- Benefits:
 - Better **understanding** of data
 - Extract **data-driven** insights
 - **Freedom** to create, format and analyse
 - **Meaningful** communication with data
 - Manage your data **sustainably**



Our Data Journey

Be Data Literate: The Data Literacy Skills Everyone Needs To Succeed. Jordan Morrow

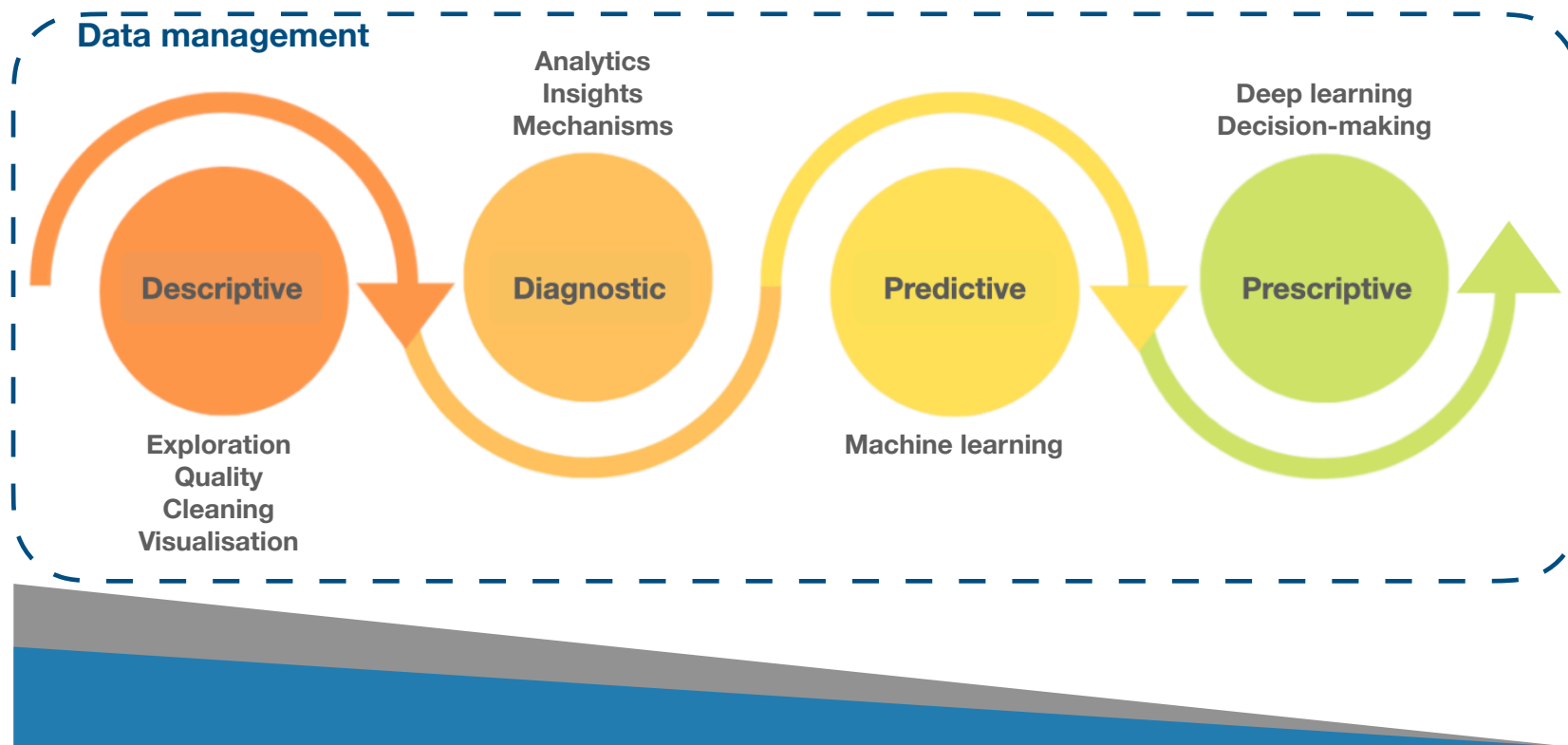
4 Levels of Data Literacy



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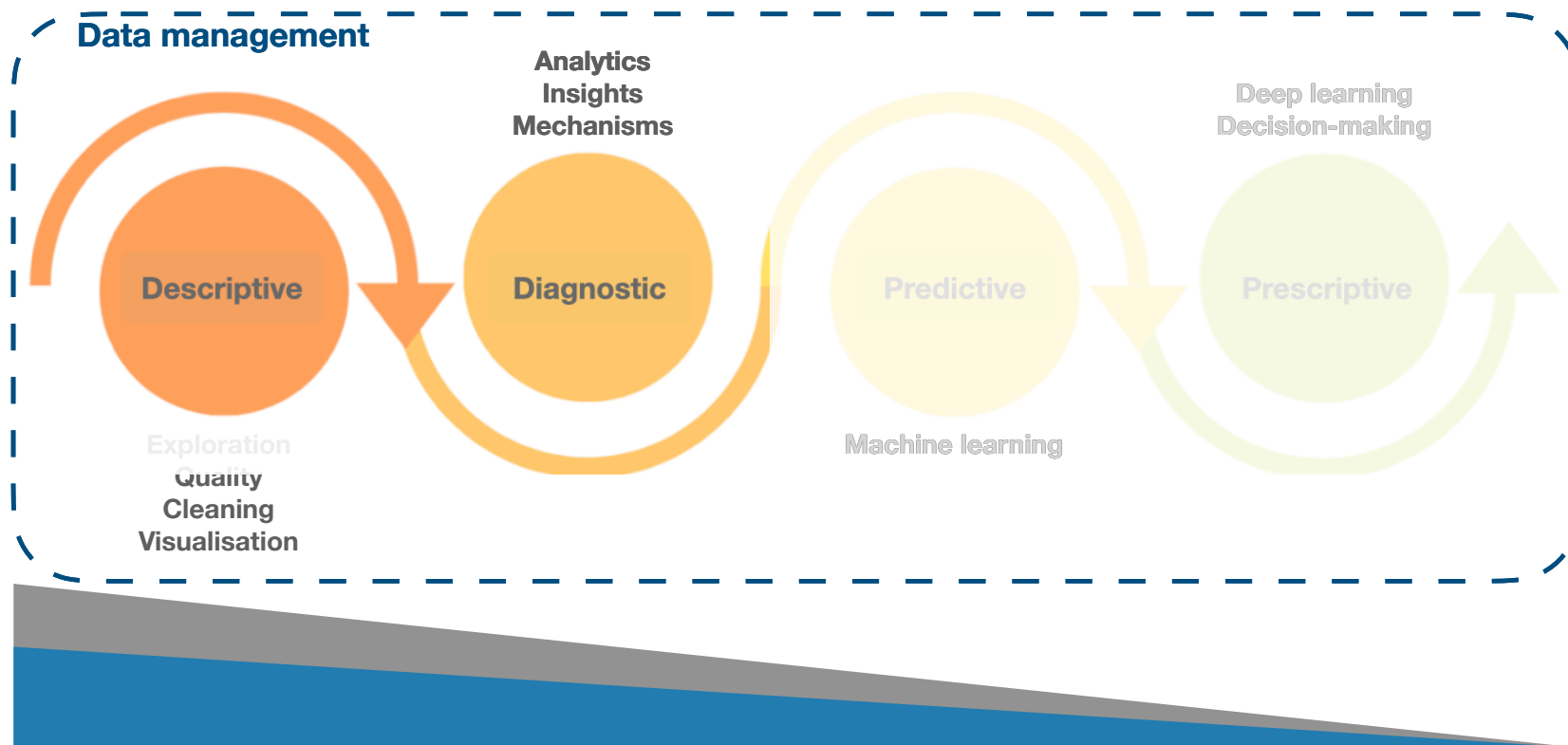
4 Levels of Data Literacy



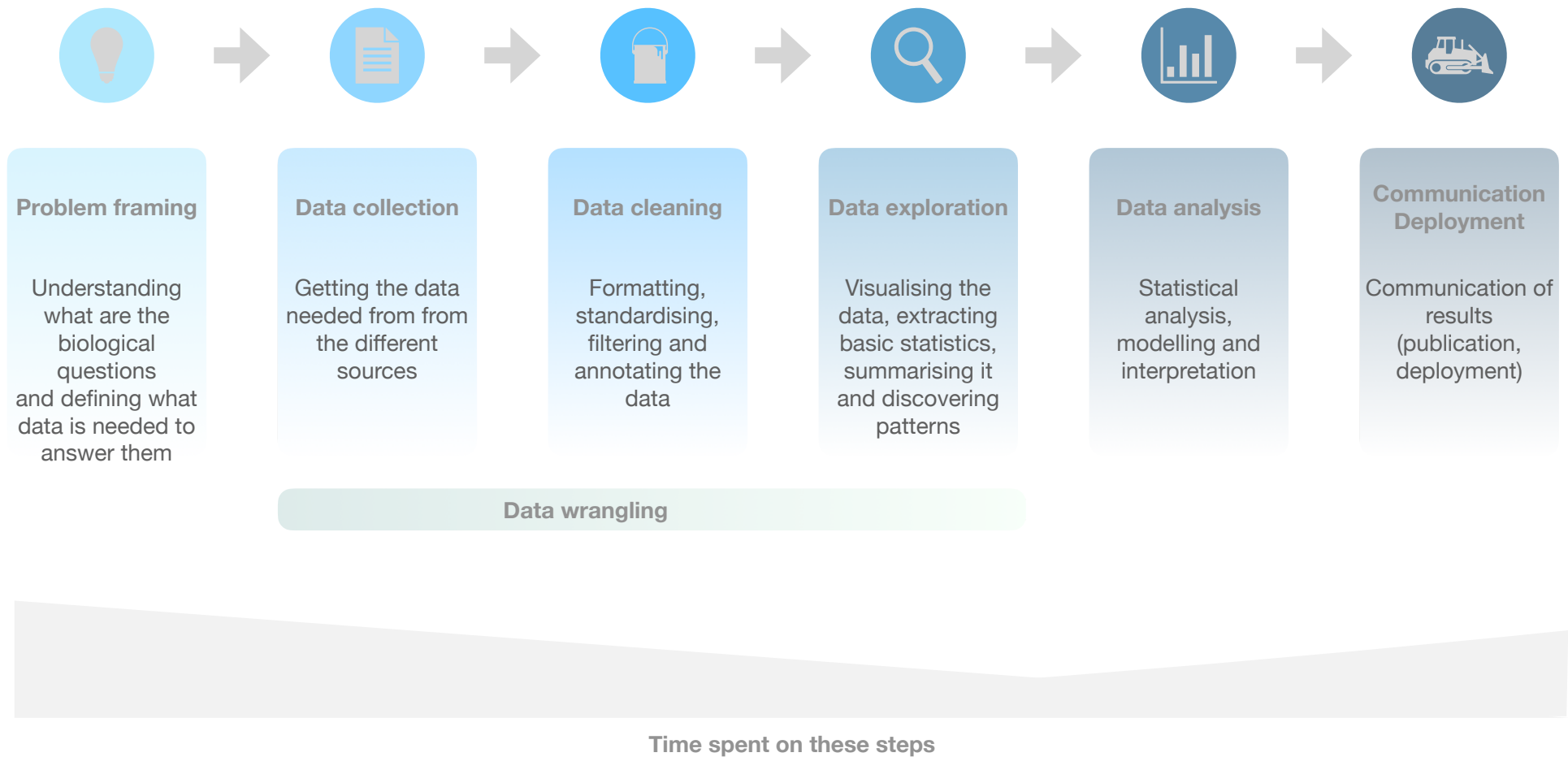
Our Data Journey

Be Data Literate: The Data Literacy Skills Everyone Needs To Succeed. Jordan Morrow

4 Levels of Data Literacy



The Data Science Process



Relevant Links

Extra readings and resources

- **Network biology: understanding the cell's functional organisation.** Albert-László Barabási & Zoltán N. Oltvai. Nature Reviews Genetics 2004 (<https://www.nature.com/articles/nrg1272>)
- **Network analysis of protein interaction data** — Online tutorial — EMBL EBI (<https://www.ebi.ac.uk/training/online/courses/network-analysis-of-protein-interaction-data-an-introduction/>)
- **Network Biology: A short introduction to the core concepts** <https://www.youtube.com/watch?v=H1bGk8PGvf8>
- **Network Visualization: a short introduction to the core concepts of network layout and clustering** <https://www.youtube.com/watch?v=OunX7ISRbgA>
- **A Guide to Conquer the Biological Network Era Using Graph Theory.** Mikaela Koutrouli, Evangelos Karatzas, David Paez-Espino, Georgios A. Pavlopoulos. Frontiers in Bioengineering and Biotechnology 2020 (<https://www.frontiersin.org/articles/10.3389/fbioe.2020.00034/full>)
- **Python for Beginners** <https://www.python.org/about/gettingstarted/>
- **How to Use Jupyter Notebook: A Beginner's Tutorial** <https://www.dataquest.io/blog/jupyter-notebook-tutorial/>
- **Getting started with Python and Jupyter Notebooks** <https://colab.research.google.com/github/jckantor/CBE30338/blob/master/docs/01.01-Getting-Started-with-Python-and-Jupyter-Notebooks.ipynb>